

Module 04: Cognition — Strategy

Learning Objectives

1. Understand that **Cognition** is thinking about the game before you move.
2. Learn about **Strategy** (Making a plan to win).
3. Know that your opponent is thinking too! (Theory of Mind).

Introduction: The Chess Master

In Chess, you don't just move pieces randomly. You think: "If I move my Knight here... She will move her Bishop there... Then I can capture the Queen!" You are playing the game in your HEAD before you play it on the BOARD. This is Cognition.

Key Concepts

1. The Plan (Future Thinking)

Strategy is a map of the future.

- "First, I will hide behind the tree."
- "Then, I will run to base when she looks away." You are predicting the future to reach your Goal.

2. Theory of Mind (Mind Reading)

To win, you have to guess what the other player is thinking.

- "He thinks I am going left, so I will go right!" (The Fake-Out).
- "She knows I have the Ace, so she will fold." You are modeling their model!

3. Decisions (The Fork in the Road)

Every turn is a choice.

- Pass or Shoot?
- Hold or Fold? Good players weigh the Risk (Losing the ball) vs. the Reward (Scoring a goal).

Activities

Activity 1: Rock, Paper, Scissors

Play 10 rounds. Try to predict what they will throw.

- "He threw Rock twice... he will probably throw Paper next."
- "She knows I know she knows..." It's a loop of thinking!

Activity 2: Tic-Tac-Toe Trap

Play Tic-Tac-Toe. Don't just try to get 3 in a row. Try to set a TRAP (Two ways to win at once). You have to think 2 steps ahead.

Summary

Strategy is the game inside the game. By thinking ahead and understanding our opponents, we turn simple rules into complex fun.

References

- *Thinking, Fast and Slow* (For Parents)
- *Elephant and Piggie: I Broke My Trunk!* by Mo Willems (Storytelling Strategy)